United International University (UIU)
Dept. of Computer Science & Engineering (CSE)
Final-Term Exam: Trimester: Fall 2024

Course Code: EEE 2123; **Course Title:** Electronics
Total Marks: 40; **Duration:** 2hr

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

There are six (6) questions in this paper. Answer all of them.

Q1. The NMOS transistor in the following Fig. has $V_t = 1$ V , Find V_g, V_d, I_D, I_G.
Here, $\mu_n C_{ox} = 0.5\, mA/V^2$, $W/L = 2$ and $I = 0.1\, mA$. **[4]**

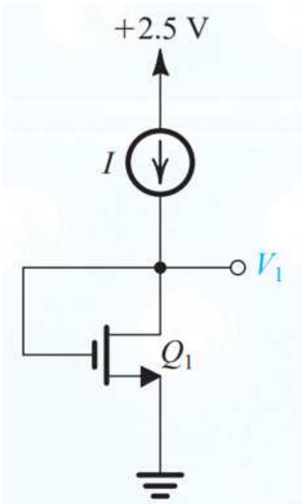

Fig 1: Circuit diagram for Q1(b)

Q2. For the circuit in Fig 2, find R_1, R_2, I_2, V_g, V_{dmin}, the largest value of $\mathbf{R_D}$ for which the transistor remains in ***saturation*** mode. The MOSFET has $V_t = 1$ V , Here
$\mu_n C_{ox} = 0.5\, mA/V^2$, $W/L = 2$. $I_D = 0.5mA$ **[8]**

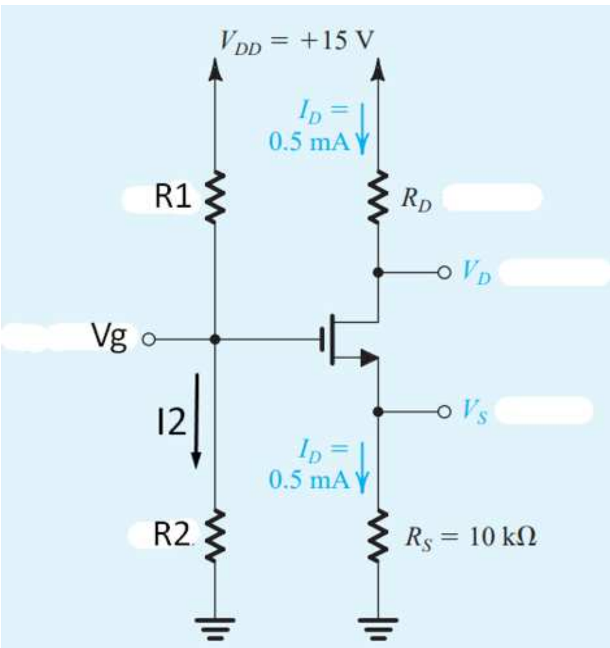

Fig 2: Circuit diagram for Q1(b)

Q3. For the circuit shown in Figure 4, the transistor parameters are RB1 = RB2 = 50KΩ, $V_{BE}(on) = 0.7$ V and $\beta = 100$. Find I_B, I_C, I_E and V_{CE}. **[7]**

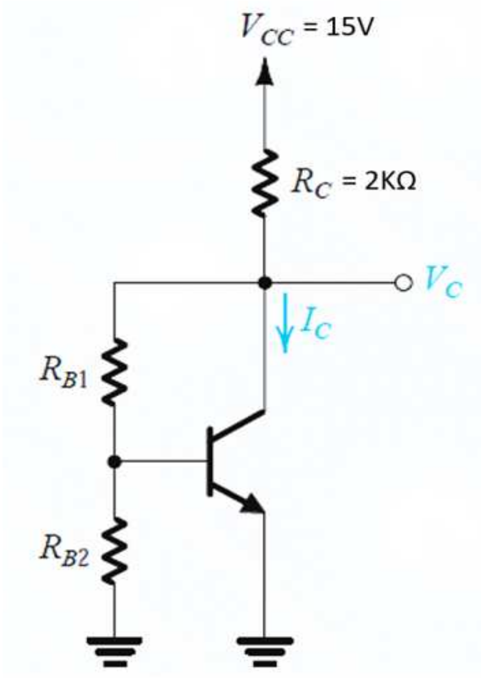

Fig 4: Circuit diagram for Q3

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4. Find all node voltages and branch currents in the following circuit in Fig. 5.
Assume, β=100 and $R_{B1} = R_{B2} =$ 250KΩ. **[7]**

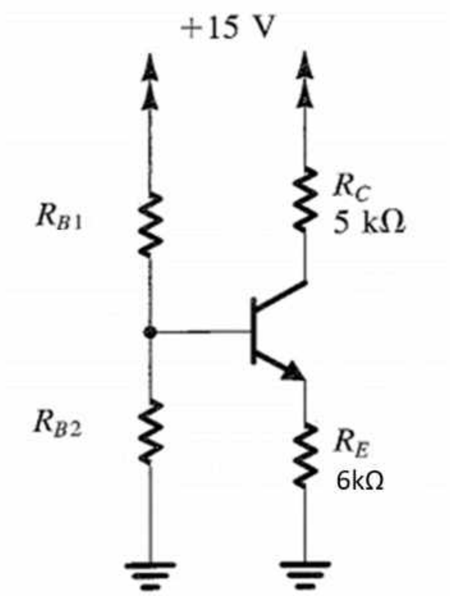

Fig 5: Circuit diagram for Q4

Q5. (a) Consider the Boolean function, $f = \overline{ABC + BC + AC + AB + \overline{B}AC}$.
Assume you are only given the inputs A, B, C; not their complements i.e. $\overline{A}$, $\overline{B}$ or $\overline{C}$.
Now design a CMOS circuit to implement the above logic with at most 5 NMOS and 5 PMOS transistors. **[5]**

(b) Explain with a CMOS inverter, why CMOS circuits have zero static power dissipation. **[2]**

Q6. Observe very carefully the input output relationship. All the blocks contain inside a single Op-Amp. Now design the circuit for the blocks if-

(a) $V_{in} = \sin(t)$ and $V_{out} = -2\cos(t)$

(b) $V_{out} = 0$ for any input.

Assume, feedback resistors (R_f) = 1k ohms.
Clearly mention all values of resistances, capacitors in both the designs. **[4+3]**

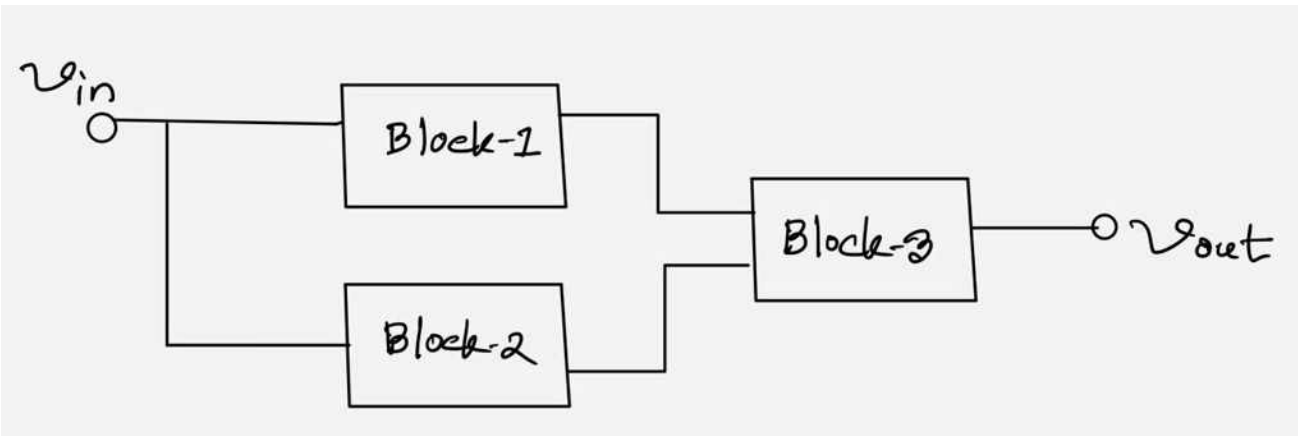

Fig 6: Circuit diagram for Q6
